Bayesian methods for clinical trials

Lecture 3: Basket trial design

Thomas Jaki & Pavel Mozgunov

MRC Biostatistics Unit

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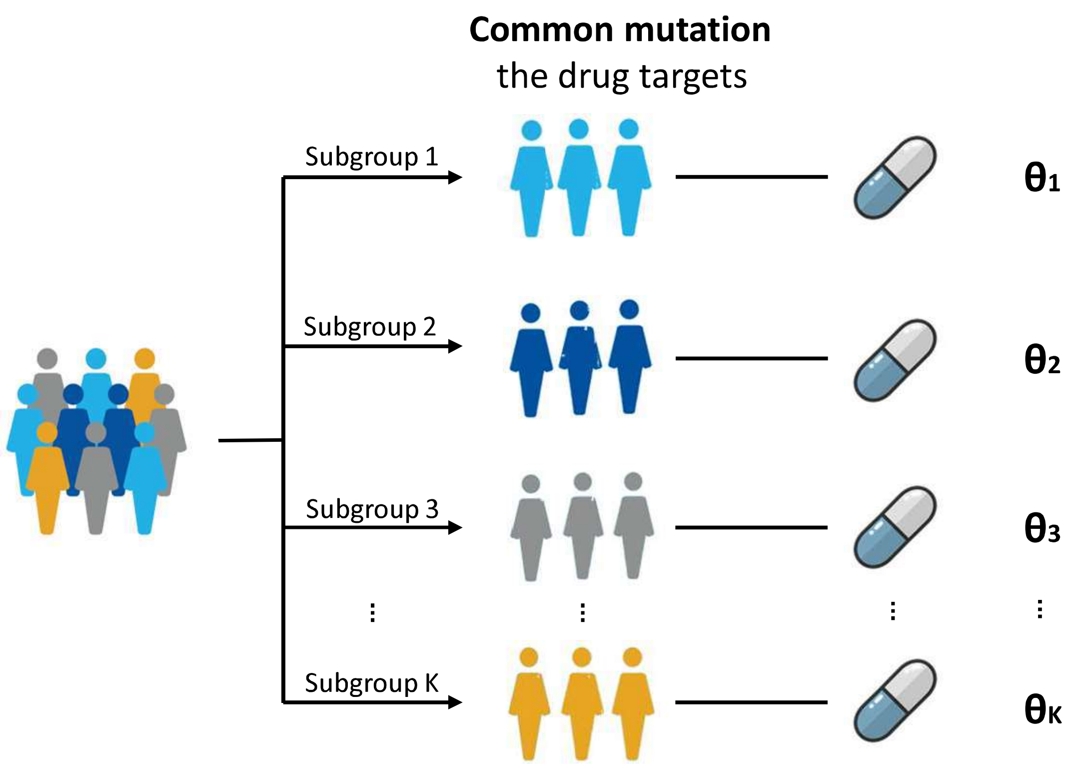
Basket trials

Setting: Common characteristic (e.g. mutation) present in multiple tumour types.

Aim: To develop **targeted therapies**

Solution Using biomarker(s) to screen patients and recruit those harbouring a common characteristic (mutation)



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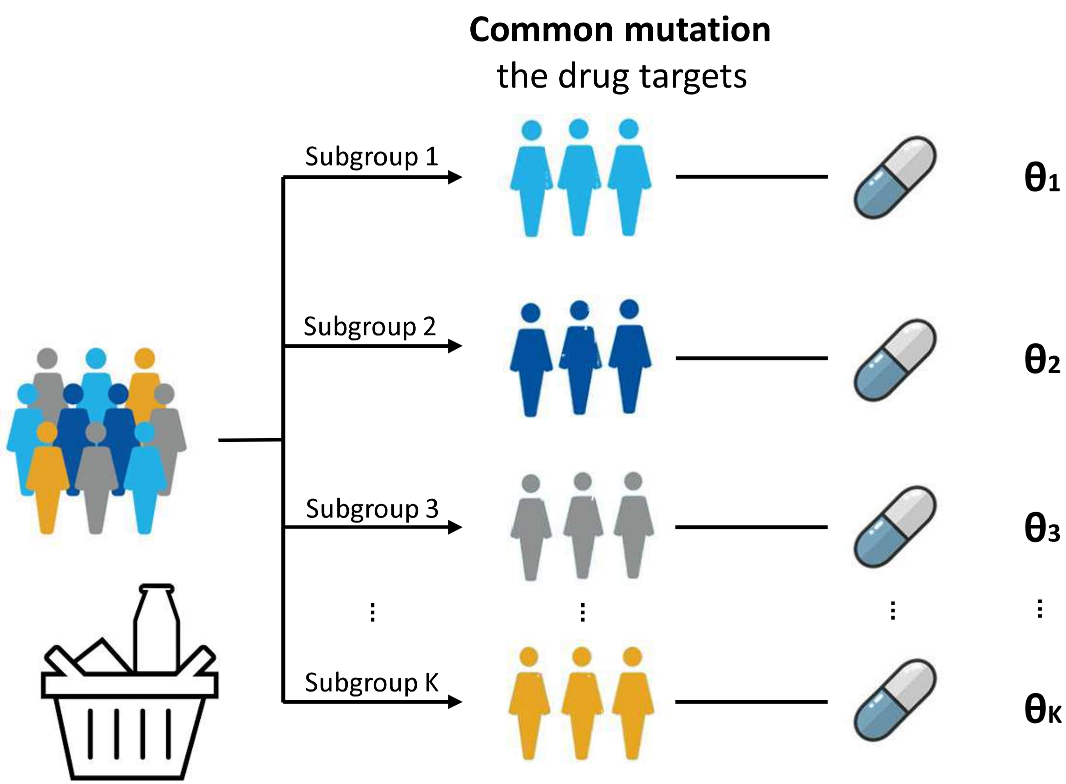

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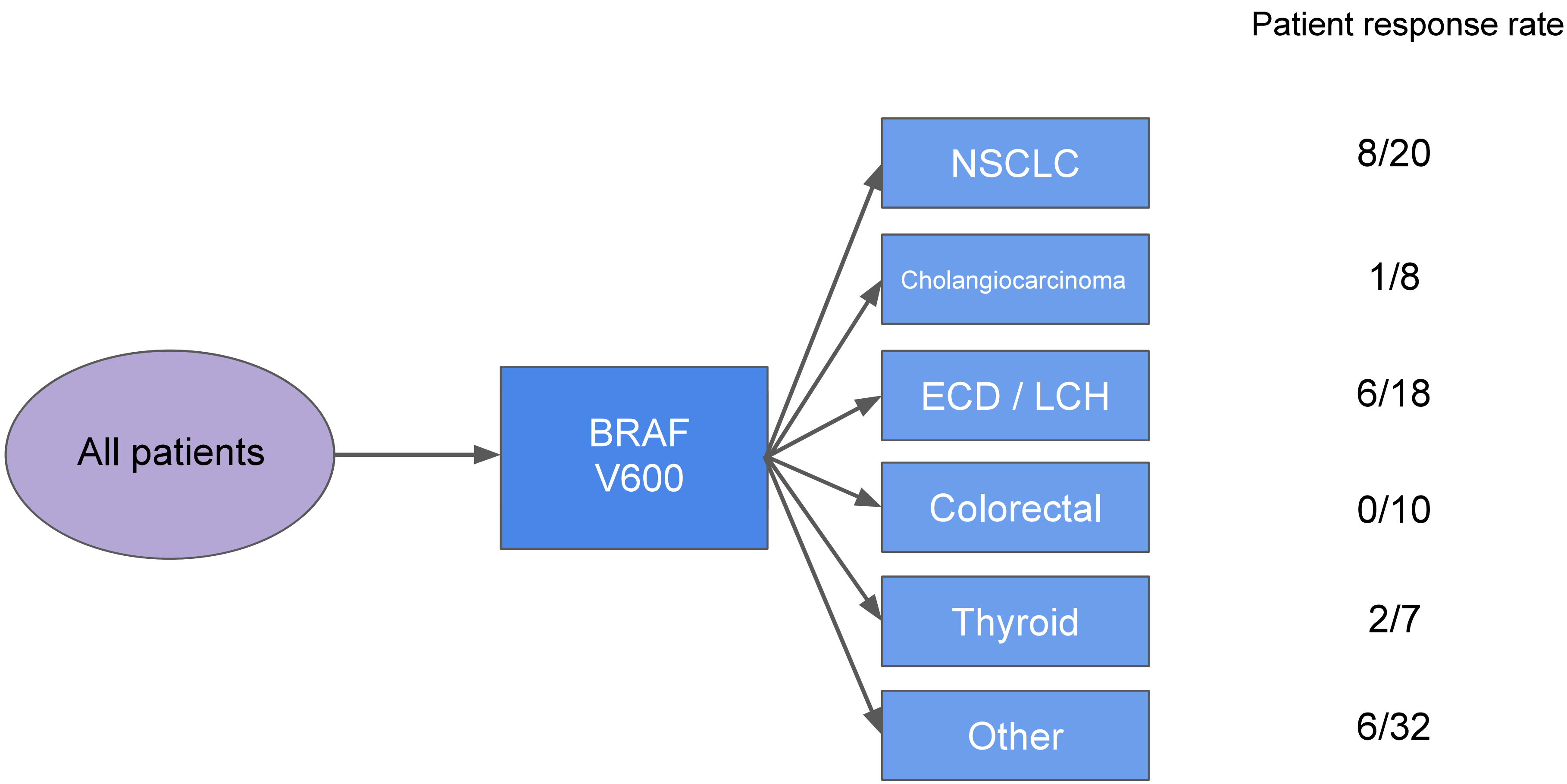
Basket trials in oncology – An example

Hyman *et al.* (2015) reported a recent basket trial, which has been designed to evaluate the efficacy of vemurafenib in BRAF-V600.

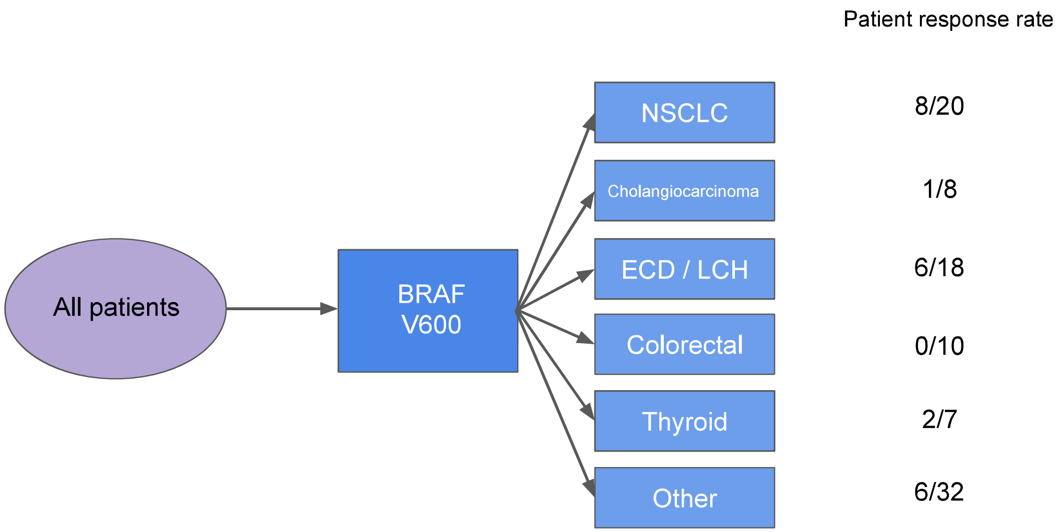
A total of 122 patients with BRAF-V600 mutations were enrolled, of which 95 entered the 6 modules.



Borrowing of information between modules

With the **common genomic mutation** targeted by the investigational drug, one may expect:

... some patient subgroups are **expected to respond similarly**.



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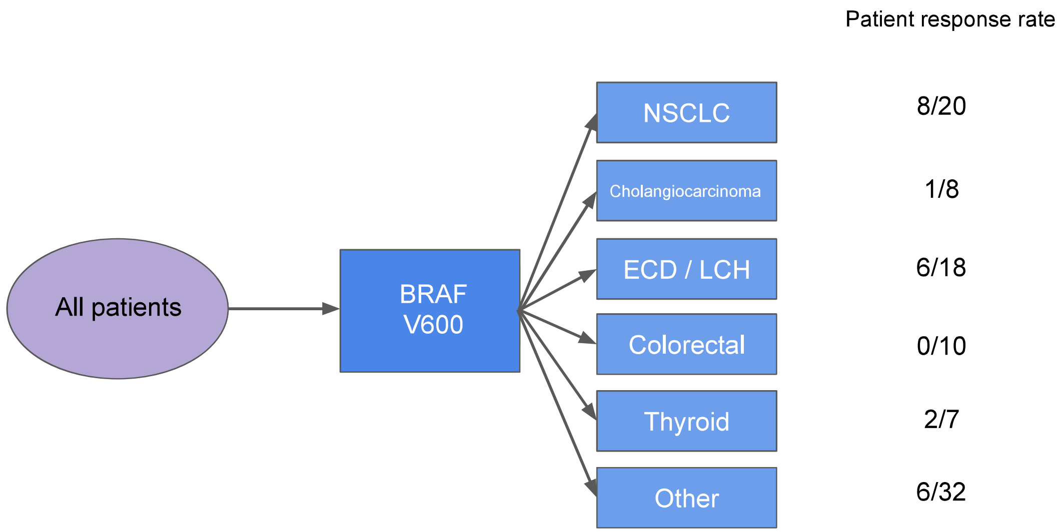

Potential analysis strategies:

- Stand-alone analyses
- Complete pooling
- Borrowing of information

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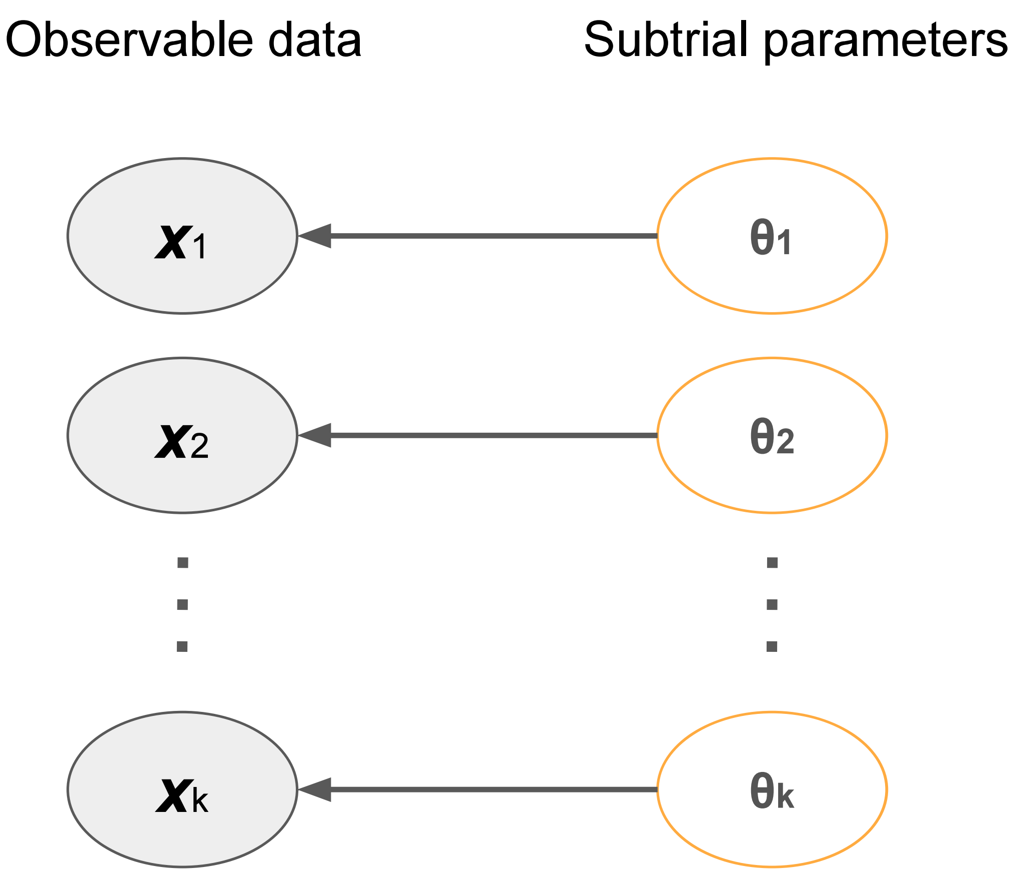
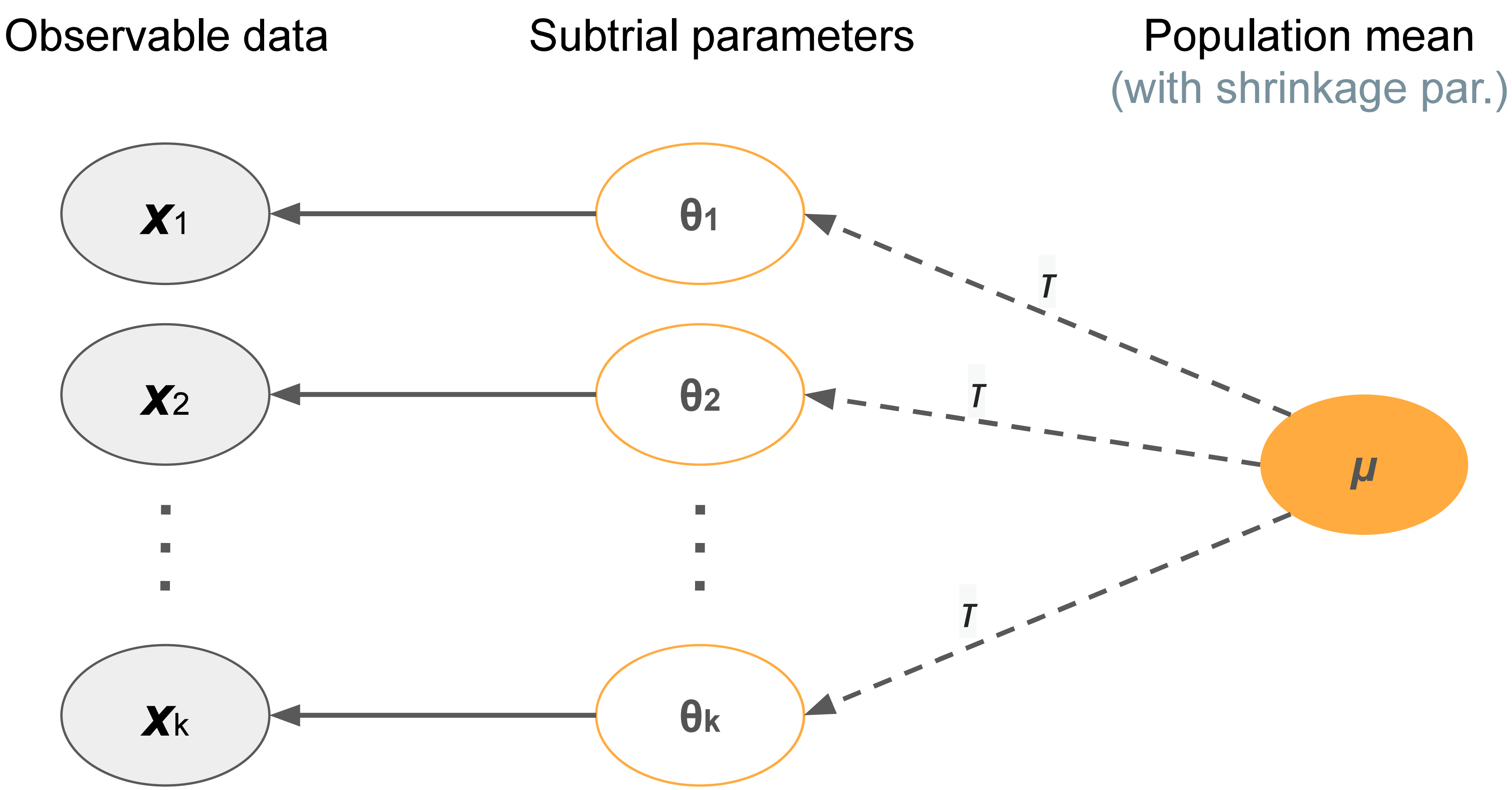
Borrowing of information



An independent analysis for NSCLC yields posterior estimate of 0.405 (see Lecture 1)

Borrowing of information



An independent analysis for NSCLC yields posterior estimate of 0.405 (see Lecture 1)

Exchangeability

The random variables $\theta_1, \theta_2, \ldots, \theta_K$ are exchangeable if

$$f_{\theta_1,\theta_2,\ldots,\theta_K}(t_1, t_2, \ldots, t_K) \overset{distr.}{=} f_{\theta_{\pi_1},\theta_{\pi_2},\ldots,\theta_{\pi_K}}(t_1, t_2, \ldots, t_K),$$

for any permutation $(\pi_1, \ldots, \pi_K)$ of the indices $\{1, 2, \ldots, K\}$.

It can be shown that

(1) i.i.d. $\Longrightarrow$ exchangeability,

(2) exchangeability $\Longrightarrow$ identically distributed.

A Bayesian hierarchical model (BHM) for binomial data:

$$
\begin{aligned}
Y_i &\sim \text{Binomial}(n_i, p_i), \qquad\qquad i = 1, \dots, k \\
\theta_i | \mu, \sigma = \log\left(\frac{p_i}{1-p_i}\right) &\sim N(\mu, \sigma^2), \\
\mu \sim N(\cdot, \cdot), &\quad \sigma \sim g(\cdot).
\end{aligned} \tag{1}
$$

- ⋆ Hierarchical modelling assumes (similarity) of the θ_is
- Borrowing occurs between all baskets $\rightarrow$ the estimates of the response rates are shrunk towards the common mean;
- Degree of shrinkage (i.e. borrowing) controlled by the so-called shrinkage/borrowing parameter, σ^2.
  - $\sigma^2 = 0 \rightarrow$ complete pooling of data from other baskets;
  - $\sigma^2 = \infty \rightarrow$ no borrowing.

Choice of hyper-prior on borrowing parameter

Based on the uncertainty of θ_i between different baskets, the following classification of values of σ was proposed

- **Small to moderate** heterogeneity: $\sigma = 0.125$ to $\sigma = 0.250$
- **Substantial to large** heterogeneity: $\sigma = 0.5$ to $\sigma = 1$

Various choices of hyper-prior $g(\cdot)$ of σ were proposed with the most common options being

- **Inverse-Gamma**

  Used in the original BHM proposal but was found to lead to a poor behaviour when σ^2 is close to 0 (Cunanan et al. 2019)

- **Half-Cauchy** (with a moderately large scale was suggested instead), e.g. Half-Cauchy$(0, 25)$ (Gelman 2006)
- **Half-Normal** with a prior standard deviation of $s = 0.5$ [95% interval on σ is (0.02, 1.12)] or $s = 1$ [95% interval on σ is (0.03, 2.24)] (Neuenschwander et al. 2016)

Now instead of analysing the first basket (8/20) alone, we conduct the analysis together with the second basket (6/18).

We use the BHM with Half-normal prior on σ^2.

The summary characteristics for the first basket:

	Independent	BHM (s = 0.5)	BHM (s = 1.0)
Mean	0.405	0.380	0.385
Median	0.402	0.377	0.381
SD	0.104	0.087	0.091
95% CI	(0.211,0.616)	(0.221,0.559)	(0.218,0.574)
Length of CI	0.404	0.338	0.357

NSCLC - hierarchical model

Assume that we were less lucky with the second basket and actually the number of responses was 0/18.

We again use the BHM with Half-normal prior on σ^2.

The summary characteristics for the first basket:

	Independent	BHM (s = 0.5)	BHM (s = 1.0)
Mean	0.405	0.321	0.362
Median	0.402	0.314	0.358
SD	0.104	0.102	0.105
95% CI	(0.211,0.616)	(0.145,0.536)	(0.172,0.579)
Length of CI	0.404	0.391	0.407

Notes

- If the baskets are **truly homogeneous**, the gains are higher if smaller variance on the borrowing parameter is assumed
- However, if there is **a heterogeneous basket**, then borrowing can lead to too much shrinkage of the estimates and even increased variance of the estimate
- The choice of the distribution of the borrowing parameters σ^2 should based on the trade-off between these two cases.

Relaxing the exchangeability assumption – EXNEX Neuenschwander et al. 2016

A Bayesian hierarchical model (BHM) assumes that the basket are exchangeable ... with probability of 1(!).

The full exchangeability assumption is often violated. To tackle this, an EXNEX model was proposed

1. EX (exchangeable component): with prior probability w_i, basket i is exchangeable and a BHM is applied.
2. NEX (nonexchangeable component): with prior probability $1 - w_i$, θ_i is nonexchangeable with any other basket and analysed independently.

$$
\begin{aligned}
Y_i &\sim \text{Binomial}(n_i, p_i), & M_{1i} &\sim \text{N}(\mu, \sigma^2), \quad \text{(EX)} \\
\theta_i &= \log\left(\frac{p_i}{1-p_i}\right), & \mu &\sim \text{N}(\cdot,\cdot), \\
\theta_i &= \delta_i M_{1i} + (1-\delta_i) M_{2i}, & \sigma &\sim g(\cdot), \\
\delta_i &\sim \text{Bernoulli}(w_i), & M_{2i} &\sim N(m_i, \nu_i). \quad \text{(NEX)} \qquad (2)
\end{aligned}
$$

Back to the example - I

Now instead of analysing the first basket (8/20) alone, we conduct the analysis together with the second basket (6/18).

We use the EXNEX with Half-normal prior on σ^2 with $s = 0.5$.

The summary characteristics for the first basket:

	Independent	BHM (s = 0.5)	EXNEX ($w = 0.5$)
Mean	0.405	0.380	0.395
Median	0.402	0.377	0.390
SD	0.104	0.087	0.097
95% CI	(0.211,0.616)	(0.221,0.559)	(0.216,0.597)
Length of CI	0.404	0.338	0.381

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Back to the example - II

Again, assume that we were less lucky with the second basket and actually the number of responses was 0/18.

We again use the EXNEX with Half-normal prior on σ^2 with $s = 0.5$.

The summary characteristics for the first basket:

	Independent	BHM (s = 0.5)	EXNEX ($w = 0.5$)
Mean	0.405	0.321	0.402
Median	0.402	0.314	0.399
SD	0.104	0.102	0.105
95% CI	(0.211,0.616)	(0.145,0.536)	(0.204,0.614)
Length of CI	0.404	0.391	0.409

Notes on EXNEX

- EXNEX provides more flexibility: information is borrowed only between baskets assigned to the EX component but not from those in the NEX component;

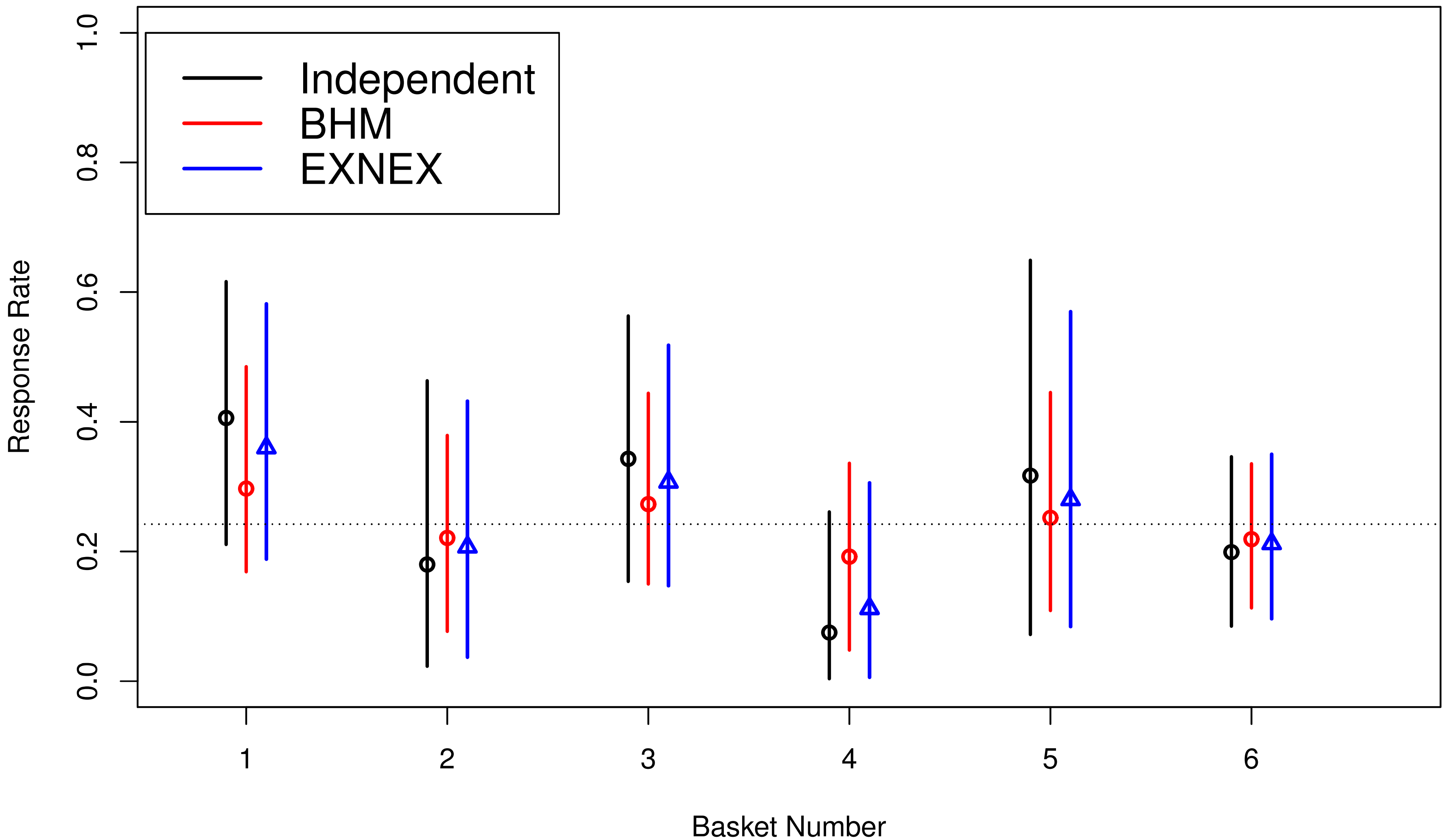
- Weight w_i is the **prior** probability reflecting how likely we believe (a-priori) that the baskets are exchangeable

- It is uncommon to have strong prior information on the probability of exchangeability, so it is suggested to fix $\pi_i = 0.5$

- This prior probability is updated to a some degree based on the homogeneity of the data but...

- The update might not be sensitive enough to the heterogeneity/ homogeneity of responses in the typically small sample sizes

Comparing different analysis models: VE-Basket



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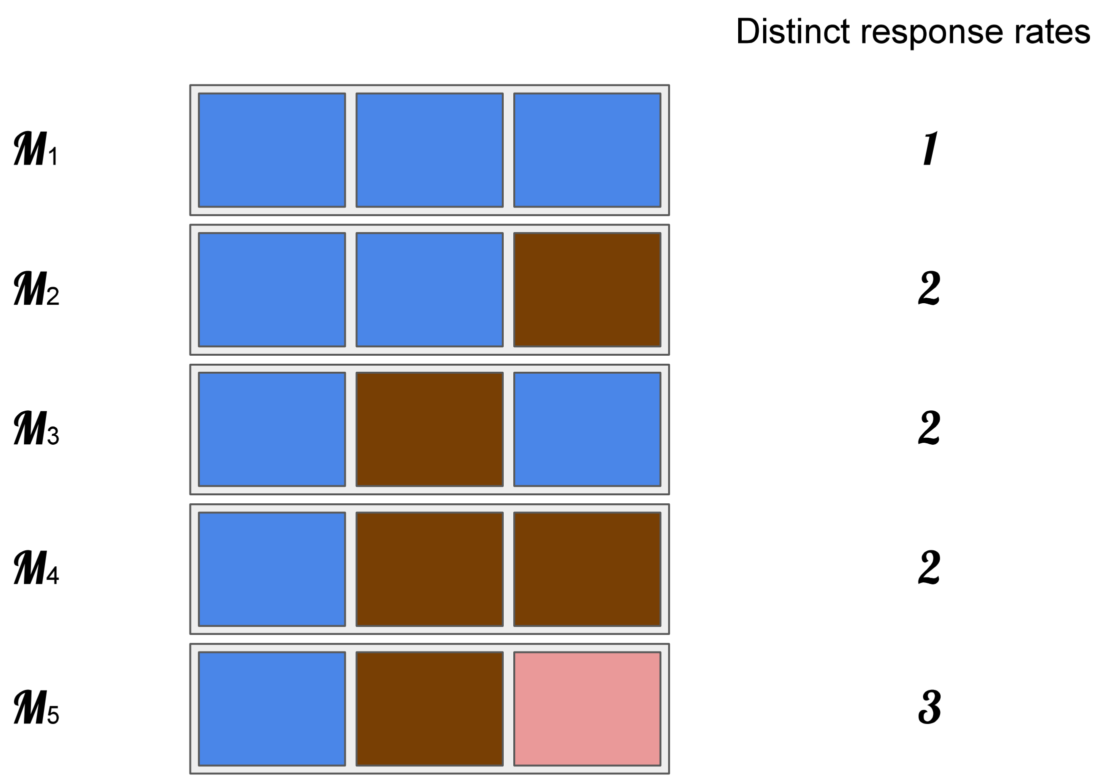
Can we do anything better?

Some modules may display more similar treatment effects between themselves than with others.

Pertinent questions to the EXNEX approach:

- ⋆ How many EX distributions are needed?
- ⋆ What is an ‘extreme subgroup’?

A model averaging approach Psioda et al. (2019)



- ⋆ Construct a *complete* model space with all possible reclassification of subgroups with identical p_i
- ⋆ Each model $\mathcal{M}_\ell$ corresponds to a distinct model stipulation

A model averaging approach Psioda et al. (2019)

Define $\mathcal{M}_j$ as model j representing a permutation of basket allocation to the EX group or NEX group.

In the EX group, results are pooled and baskets have one shared response rate.

A weakly informative Beta prior is placed on the response rates, while a prior on each model, $f(\mathcal{M}_j)$, is also required. The posteriors $f(p_i|\mathcal{M}_j)$ and $f(\mathcal{M}_j|D)$ are computed.

BMA procedure to obtain a summary statistic for basket k. For example, for the probability of being above 0.25

$$\mathbb{P}(p_i > 0.25|\mathrm{Data},\ \mathrm{Prior}) = \sum_j \mathbb{P}(p_i > 0.25|\mathcal{M}_j, D) f(\mathcal{M}_j|D)$$

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Comments

- Many more models proposed
- Often single arm baskets considered, but randomized versions (borrowing on contrast parameter) available
- Often binary outcomes considered
- Basket structure in trials becoming more common, yet borrowing as main analysis still rare

Regulatory challenges

- Is exchangeability clinically/mechanistically plausibe
  - Any borrowing model will borrow if response is similar
- Is information in a basket dominated by borrowed information
- Control of false decision error

- Decision are made based on the **posterior probabilities for the treatment effect** given the clinical trial data;

For example, in a single-arm trial, one is claiming **benefit** if

$$\mathrm{Prob}\left[\theta \geq p_0 \mid \mathrm{Data},\ \mathrm{Prior}\right] > c$$

where

- p_0 is an effect threshold
- c is probability threshold (often chosen to control type I error)
- 'Data" is the observed trial data
- "Prior" is specified prior distribution on the treatment effect.

Bayesian decision-making: example

The type I error is then defined as

$$\mathrm{Prob}\{\mathrm{Prob}\,[\theta \geq p_0 \mid \mathrm{Data},\ \mathrm{Prior}] > c|\theta = p_0\}$$

The power is defined as

$$\mathrm{Prob}\{\mathrm{Prob}\,[\theta \geq p_0 \mid \mathrm{Data},\ \mathrm{Prior}] > c|\theta = p_1\}$$

Null configuration

- What would be the null hypothesis in a basket trial setting?
- With K baskets, one can consider the configuration $p_1 = p_2 = \ldots = p_k = p_0$ where p_0 is a null response rate for the calibration of the probability threshold c.
- This assumes that all (!) baskets are homogeneous. What would one expect to see if the some of the baskets has interesting treatment effect and some null?
- Were one to calibrate under the case of the maximum type I error, there will be no more advantages of using a basket trial design (Kopp-Schneider et al. 2020)

Robust Calibration Procedure (RCaP) Daniells et al. (2025)

- Typically type I error controlled by tuning c using simulations under the global null
- RCaP
  - Defines several nulls of interest
  - Calibrate c such that type I error is controlled *on average* across selected null cases

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